

August 19, 2025

```
[11]: import torch
      from torch import nn
```

0.0.1

```
[12]: def corr2d(x, k):
      h, w = k.shape
      y = torch.zeros((x.shape[0] - h + 1, x.shape[-1] - w + 1))
      for i in range(y.shape[0]):
          for j in range(y.shape[1]):
              y[i, j] = (x[i:i+h, j:j+w] * k).sum()
      return y
```

```
[13]: x = torch.tensor([[0, 1, 2], [3, 4, 5], [6, 7, 8]])
      k = torch.tensor([[0, 1], [2, 3]])
      corr2d(x, k)
```

```
[13]: tensor([[19., 25.],
            [37., 43.]])
```

```
[ ]: class Conv2d(nn.Module):
      def __init__(self, kernel_size):
          super().__init__()
          self.weight = nn.Parameter(torch.rand(kernel_size))
          self.bias = nn.Parameter(torch.zeros(1))

      def forward(self, x):
          return corr2d(x, self.weight) + self.bias
```

```
[16]: x = torch.ones((6, 8))
      x[:, 2: 6] = 0
      k = torch.tensor([[1, -1]])
      y = corr2d(x, k)
      y
```

```
[16]: tensor([[ 0.,  1.,  0.,  0.,  0., -1.,  0.],
          [ 0.,  1.,  0.,  0.,  0., -1.,  0.],
          [ 0.,  1.,  0.,  0.,  0., -1.,  0.]])
```

```
[ 0.,  1.,  0.,  0.,  0., -1.,  0.],
[ 0.,  1.,  0.,  0.,  0., -1.,  0.],
[ 0.,  1.,  0.,  0.,  0., -1.,  0.]])
```

0.0.2

```
[18]: conv2d = nn.Conv2d(1, 1, kernel_size=(1, 2), bias=False)

x = x.reshape((1, 1, 6, 8))
y = y.reshape((1, 1, 6, 7))

for i in range(10):
    y_hat = conv2d(x)
    l = (y_hat - y) ** 2
    conv2d.zero_grad()
    l.sum().backward()
    conv2d.weight.data -= 0.01 * conv2d.weight.grad
    if (i + 1) % 2 == 0:
        print(f'epoch {i + 1}, loss {l.sum():.3f}, weight {conv2d.weight.
↳reshape(-1)}')
```

```
epoch 2, loss 4.164, weight tensor([ 0.4866, -0.4776], grad_fn=<ViewBackward0>)
epoch 4, loss 2.493, weight tensor([ 0.5997, -0.5982], grad_fn=<ViewBackward0>)
epoch 6, loss 1.495, weight tensor([ 0.6895, -0.6893], grad_fn=<ViewBackward0>)
epoch 8, loss 0.896, weight tensor([ 0.7595, -0.7595], grad_fn=<ViewBackward0>)
epoch 10, loss 0.538, weight tensor([ 0.8138, -0.8137], grad_fn=<ViewBackward0>)
```